Rough sea surface in the discrete nonlocal boundary condition of the rational parabolic equation: analytic continuation of the Miller–Brown reflection coefficient

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Abstract

The discrete nonlocal boundary condition (NLBC) of the split-step Padé parabolic equation accounts for an arbitrary angle-dependent reflection coefficient of the lower boundary, provided the coefficient admits an analytic continuation to the complex vertical wavenumbers sampled by the inverse z-transform. We extend the approach to a rough sea surface described by the Miller–Brown coherent reflection factor. Its continuation is entire but grows off the real axis, so that $1 + R$ acquires zeros inside the contour and the boundary condition becomes unstable; no continuation that is bounded by one can reproduce the factor near grazing incidence. Two remedies are developed: lifting the boundary to a reference plane above the mean level, and a zero-free fitted coefficient $1 + \tilde{R} = (1 + R_f)e^K$ combined with a small lift. Long-range tests over ducts reveal the conditions such a coefficient must satisfy in addition to the absence of zeros: passivity for all propagating waves, passivity in the trapping range of the atmosphere, and accuracy on the complex wavenumbers of range-decaying modes. Beyond the radio horizon Miller–Brown raises the field by tens of decibels. An independent modal and branch-cut solution shows that this is the lateral wave that the model itself produces at grazing incidence, because its effective surface impedance is odd in the vertical wavenumber; it is not an artefact of the boundary condition. It contradicts the grazing behaviour established by Barrick, that the roughness-modified impedance of any rough surface tends to a constant. We therefore add a constant grazing impedance: the roughness admittance of Miller–Brown is kept above a blend angle and frozen below it, evenly in the vertical wavenumber. With the fitted coefficient matched to first order at grazing, the parabolic equation reproduces the creeping-mode attenuation of this impedance to 0.01 dB/km, and the beyond-horizon excess disappears, except where the blend angle falls inside the range of the creeping modes (X band, 15 m/s), where it is reduced from 40 to 11 dB. The method agrees with an exact plane-wave-spectrum solution to 0.1–1.2 % and is demonstrated for L, S and X band, wind speeds up to 15 m/s and three refractivity profiles over 60 km.

1 Introduction

The parabolic equation (PE) is the standard tool for tropospheric radio-wave propagation over the sea. The surface enters through the lower boundary condition. The usual Leontovich impedance condition assumes a grazing-angle independent impedance, and the split-step Fourier method with the mixed Fourier transform inherits this restriction. In [1] it was shown that the discrete nonlocal boundary condition (NLBC) of the finite-difference split-step Padé PE accommodates an *arbitrary* Fresnel reflection coefficient $R(k_z)$, including the Brewster-angle zero of vertical polarisation. The condition is that R admits an analytic continuation to the complex vertical wavenumbers sampled by the inverse z-transform. The conclusion of [1] named the Miller–Brown rough-surface model [2] as a natural next step and identified the analytic continuation as the open problem. Miller–Brown is the coherent reflection model most widely used with split-step Fourier PE codes; its accuracy for forward propagation was assessed numerically in [7].

Related work. Barrick [8] criticised the use of Miller–Brown in split-step codes: the resulting impedance and admittance depend strongly on the grazing angle, whereas exact modal solutions give a roughness-modified impedance that tends to a constant at grazing, for both polarisations and any roughness. Perturbation theory [9] and the Dyson-equation treatment of the coherent field [10] lead to the same structure, an effective impedance equal to the smooth one plus a roughness term, and the constant has been extracted numerically from fields computed over rough surfaces [11]. Shadowing corrections to the Ament and Miller–Brown factors reduce the effective roughness near grazing [12, 13, 14]. The small-slope approximation [15] covers the Kirchhoff and Bragg regimes in one theory but loses validity once the grazing angle falls below the rms slope. A different route avoids a reflection model altogether: random sea surfaces generated from a wave spectrum (Pierson–Moskowitz [16], JONSWAP [17], or Elfouhaily [18], which uses JONSWAP for the long waves) are propagated over as terrain and averaged [19, 20, 21], or the spectrum is split, its long waves forming the geometry and its short waves a roughness coefficient [22].

This report addresses that problem. Section 2 recalls the NLBC and the roughness models. Section 3 shows why the plain continuation fails and why no bounded continuation can be exact. Section 4 develops a lifted reference plane and a zero-free fitted coefficient, together with the stability conditions found on long-range problems. Section 5 analyses the field beyond the radio horizon, and Section 6 replaces the grazing extrapolation of Miller–Brown by a constant impedance. Validation and a long-range demonstration follow in Section 7.

2 Nonlocal boundary condition and rough-surface reflection

The reduced field $u = \psi e^{-ikx}$ of the two-dimensional Helmholtz equation is marched in range by a rational (Padé) approximation of the one-way propagator,

$$u^{n+1} = \prod_{l=1}^{p} \frac{1 + a_l L}{1 + b_l L}\, u^n, \qquad L = \frac{1}{k^2}\frac{\partial^2}{\partial z^2} + \left(m^2 - 1\right), \tag{1}$$

each factor being solved by a fourth-order Numerov scheme in z [5, 1]. Near the lower boundary the medium is homogeneous and the discrete vertical equation has solutions $\mu^{\pm j}$, $|\mu| < 1$, with $k_z = \log\mu/(i\Delta z)$. A reflecting surface relates the down- and up-going parts by $R(k_z)$, giving the z-transformed boundary relation

$$\frac{\chi_1}{\chi_0} = \mu + \frac{\mu^{-1} - \mu}{1 + R(k_z)}, \tag{2}$$

evaluated for each eigenvalue of the Padé pencil. The convolution coefficients D_m of the NLBC follow from an inverse z-transform on the circle $|\xi| = \tau > 1$, computed by adaptive Filon–Clenshaw–Curtis quadrature.

Roughness models. With $x = 2(\sigma k_z)^2$, σ the rms wave height, the coherent reflection coefficient about the mean level is $R = R_f\,\rho$, where R_f is the Fresnel coefficient and

$$\rho_{\mathrm{MB}} = e^{-x} I_0(x) \quad \text{(Miller–Brown [2])}, \qquad \rho_{\mathrm{A}} = e^{-x} \quad \text{(Ament [3])}. \tag{3}$$

For a fully developed sea $\sigma = 0.0051\,U^2$ (U the wind speed in m/s) [4].

3 The analytic-continuation problem

Where R is sampled. The circle $|\xi| = \tau$ maps onto a closed loop in the first quadrant of the k_z plane (Fig. 1). It runs along the real (propagating) axis, up the imaginary (evanescent) axis and back along an arc; its interior is the image of $|\xi| > \tau$. The boundary ratio is analytic there, and the D_m are causal and stable, if and only if R is analytic inside the loop and $1 + R$ has no zeros there.

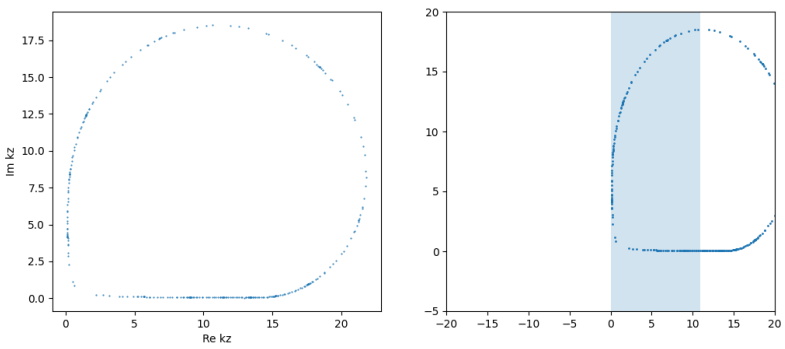


Figure 1: Vertical wavenumbers sampled by the inverse z-transform at the lower boundary (3 GHz, Padé [5/6]). Right: zoom near the origin; shaded: the propagating range $[0, k_{z,\max}]$.

Failure of the plain continuation. Both roughness factors are entire functions of k_z, but they grow like $e^{4(\sigma \operatorname{Im} k_z)^2}$ off the real axis. For the H-polarised sea, $R_f \approx -1$ and the zeros of $1 + R$ are close to those of $1 - \rho_{\mathrm{MB}}(x)$: the first lies at $|x| \approx 3.5$, $\arg x \approx 102°$, i.e. at $|k_z| \approx 1.32/\sigma$, $\arg k_z \approx 51°$. It falls inside the loop for any appreciable roughness (for $\sigma = 0.3$ m at 3 GHz the first zeros are at $k_z \approx 2.78 + 3.42i$, and the loop reaches $|k_z| \approx 22$). The resulting boundary condition is wrong: the field error against the exact solution of Section 7 is 45 % for $\sigma = 0.3$ m, H-pol.

No exact bounded continuation. A coefficient $\tilde{R}$ that is analytic and bounded by one inside the loop cannot equal $R_f \rho_{\mathrm{MB}}$ on a segment of the real axis. By the identity theorem it would coincide with the entire continuation, which is unbounded. Approximation does not escape either: requiring $|\tilde{\rho}| \le 1$ on the evanescent axis, where the continuation $\rho_{\mathrm{MB}}(-|x|) = e^{|x|} I_0(|x|)$ exceeds one, forces a jump in the argument of the boundary data at $x = 0$. Its harmonic conjugate is a phase of order $(x/\pi)\log(1/x)$ near grazing (a Kramers–Kronig constraint). Numerically, the best bounded approximation of ρ_{MB} carries 2–18° of phase.

4 Remedies

4.1 Reference plane above the mean level

A boundary condition can only represent scatterers below it, while the wave crests rise above the mean level. Placing the boundary at height z_0 and re-referencing the reflection coefficient,

$$R_{z_0}(k_z) = R_f(k_z)\, \rho(k_z)\, e^{2ik_z z_0}, \tag{4}$$

reproduces the mean-level rough reflection exactly at every real grazing angle (a pure reference-plane phase), while $|e^{2ik_z z_0}| = e^{-2z_0 \operatorname{Im} k_z}$ damps the growth. The smallest admissible lift follows from the maximum-modulus principle: if $|R_{z_0}| \le 1$ on the loop, then $|R_{z_0}| < 1$ inside it, and $1 + R$ cannot vanish. Since $\log |R_{z_0}|$ is linear in z_0 at every contour point,

$$z_0 = \max_{\text{loop}} \frac{\log |R_f \rho|}{2 \operatorname{Im} k_z}. \tag{5}$$

The method is exact but z_0 grows like σ^2 times the loop size: 2.6 m for $\sigma = 0.3$ m at 3 GHz, but 11 m and 78 m for X band at 10 and 15 m/s. The field below z_0 is not computed, and a large lift removes refracting air below the plane.

4.2 Zero-free fitted coefficient

The stability condition does not require $|\tilde{R}| \le 1$, only that $1 + \tilde{R}$ have no zeros. This is built in by

$$1 + \tilde{R}(k_z) = \big(1 + R_f(k_z)\big)\, e^{K(k_z)}, \tag{6}$$

since $1 + R_f$ is zero-free (the smooth condition is stable) for *any* analytic K. K is fitted by least squares to $\log\big[(1+R)/(1+R_f)\big]$:

- *Basis.* A polynomial in k_z of degree 40 (80 if needed), with the Vandermonde-with-Arnoldi orthogonalisation [6]. When a lift is used, $\log(1+R_{z_0})$ is a power series in the lift phase, so $K = \sum_{n\le 6} e^{2ink_z z_0} P_n(k_z)$. Each harmonic is bounded by one in the upper half plane.
- *Accuracy* on the propagating range $[0, 1.2\,k_{z,\max}]$, measured as the relative error of $1 + R$. Near grazing $R \to -1$, and an absolute error in R would misplace the effective surface impedance.
- *Regularisation.* A Tikhonov term on the loop is relaxed until the tolerance (5×10^{-3}) is met, keeping $|\mathrm{Re}\, K| \le 15$. The loop is thinned on a spatial grid: it holds tens of thousands of nearly coincident samples, and a stride aliases with their ordering.

For strong roughness the zeros of the continued $1 + R$ approach the propagating range and no moderate K fits. The boundary is then lifted by the smallest z_0 (in fractions of σ) that makes the fit admissible. In all cases studied this is a few centimetres to 1.7 m, compared with up to 78 m for the pure lift.

4.3 Stability conditions found on long-range problems

Zero-freeness proved necessary but not sufficient. Each of the following conditions was found from a failure on a long-range problem, and each is now part of the admissibility test of a fit:

1. *Passivity for propagating waves:* $|\tilde{R}| \le 1$ on the whole real stretch of the loop, not only on the fitted range. Otherwise a propagating wave gains energy at every bounce and diverges in a duct (L band, 15 m/s, evaporation duct).
2. *Passivity in the trapping range:* $|\tilde{R}| \le 1$ in the quarter disc $|k_z| \le k\sqrt{2 \cdot 10^{-6}\Delta M}$ above the real axis, where ΔM is the modified-refractivity deficit of a trapping layer. A layer returns these waves to the surface, and an active surface would resonate with it (X band, 15 m/s).
3. *Accuracy for range-decaying modes:* the creeping modes beyond the horizon sit at $\arg k_z \approx -60°$, $|k_z| = \sqrt{|a_n|}\,(k^2 g)^{1/3}$ with a_n the Airy zeros and $g = \mathrm{d}m^2/\mathrm{d}z$. Trapped duct modes lie just below the real axis. K is fitted to the continued R on these sectors as well; otherwise the polynomial is unconstrained there. With the sectors included, the creeping-mode attenuations computed with $\tilde{R}$ agree with those of the exact Miller–Brown coefficient to 0.01 dB/km.
4. *First-order accuracy at grazing:* K is matched to the value and the slope of its target at $k_z = 0$ (central differences across the origin, as heavily weighted rows of the least-squares problem). Since $1 + R_f = O(k_z)$, then $\tilde{R} - R = O(k_z^3)$ and the effective impedance (Section 5) is exact to $O(k_z^2)$: the fit adds no odd linear term to it. Without this condition a residual slope of K of order 10^{-3}, harmless for Miller–Brown, left a lateral wave some 20 dB below that of Miller–Brown when the grazing impedance is small (Section 6). With it, the odd part of the fitted impedance near grazing drops from 10^{-5} to 10^{-10} of its value.

5 The field beyond the radio horizon

Over a rough sea described by Miller–Brown the computed field far beyond the horizon in a standard atmosphere exceeds the smooth-sea field by tens of decibels at 10–15 m/s. It is nearly uniform in height and decays slowly with range. An independent solution of the continuous problem shows that this is a property of the roughness model, not of the boundary condition. We found no support for it in other studies: PE computations with shadowed roughness factors or explicit rough surfaces report propagation loss that increases with wind speed [19, 13], and the effective impedance of a rough surface is constant at grazing [8], whereas that of Miller–Brown is not.

Table 1: Field relative to the smooth sea (dB) beyond the horizon, X band, standard atmosphere, antenna at 20 m, 58 km range, 10 m height.

Wind speed	Independent, exact MB	Independent, MB at $z_0 = 3\sigma$	PE, fitted $\tilde{R}$
5 m/s	−4.2	+2.5	+7.0
10 m/s	+26.2	+26.0	+26.4
15 m/s	+40.6	+38.1	+39.7

Independent solution. In the earth-flattened standard atmosphere $m^2 = 1 + gz$, the vertical solutions are Airy functions. The line-source Green's function is

$$G(x, z) = \frac{1}{2\pi} \int g(z, z_s; k_x)\, e^{ik_x x}\, dk_x, \tag{7}$$

with $g = \phi_b(z_<)\phi_{\text{out}}(z_>)/W$. Here $\phi_{\text{out}} = \text{Ai}(s) - i\,\text{Bi}(s)$ radiates upward, and ϕ_b satisfies the impedance condition $u'/u = i\kappa$ with

$$\kappa(k_z) = k_z \frac{R(k_z) - 1}{R(k_z) + 1}. \tag{8}$$

Deforming the contour for large x gives the residues at the creeping modes plus an integral along the branch cut at $k_x = k$ ($k_z = 0$). The two sides of the cut correspond to $\pm k_z$, and the cut integral (the lateral wave) is driven by the odd part of κ. For the smooth sea (H-pol) $\kappa = -q$, $q = k\sqrt{\varepsilon - 1 + k_z^2/k^2}$, is even: no lateral wave, and the field decays exponentially (2.41 dB/km at X band, against 2.50 dB/km from the PE). With Miller–Brown, near grazing

$$\kappa \approx -\frac{q}{1 + q\sigma^2 k_z}, \tag{9}$$

which is odd in k_z. The model itself predicts a lateral wave. More generally, κ of $R = R_f\rho$ is even if and only if $\rho(k_z)\rho(-k_z) = 1$, which an even roughness factor violates. (With the down-going wave $e^{-ik_z z} + Re^{ik_z z}$ the condition reads $u_z = i\kappa u$; real negative κ is absorbing.)

Figure 2 shows $|\kappa|$ against the grazing angle. For the smooth sea it is constant. With Miller–Brown or Ament at H polarisation it falls roughly as $1/k_z$ over two decades of angle and has not reached the smooth value even at 0.001° at 15 m/s; the constant region of the model, $k_z \ll 1/(q\sigma^2)$, lies below 10^{-4} rad at 3 GHz and 10 m/s. Ament with Smith shadowing moves the transition to angles about ten times smaller but does not remove it. At V polarisation all models stay near the smooth value below about 0.05°, in line with Barrick's observation that Miller–Brown is roughly adequate for VV but not for HH [8].

Comparison. Table 1 compares the excess field over the smooth sea at 58 km, 10 m height. The PE with the fitted coefficient follows the independent Miller–Brown solution at 10 m/s and above. At 5 m/s, deep in the shadow (path loss about 235 dB), it differs by about 11 dB: the continued impedance has a singularity at $k_z = -1/(q\sigma^2)$, close to the origin, which a polynomial K cannot represent. The lateral wave samples R on that side of the cut.

What does not remove it. Lifting the reference plane leaves the excess essentially unchanged (Table 1), because the lift factor $e^{2ik_z z_0}$ is itself odd across the cut. A passive roughness factor $|\tilde{\rho}| \le 1$, built as the Schwarz integral of $\log|\tilde{\rho}| = \log\rho_{\text{MB}}$ on the propagating axis, introduces an $x \log x$ branch point at grazing. That again creates a lateral wave, and unless $\tilde{\rho}(0) = 1$ is enforced exactly, it also moves the trivial grazing root of the mode equation off zero into a lossless mode. Finally, an exactly lateral-wave-free factor $\tilde{\rho} = e^{k_z H(k_z^2)}$ cannot follow the quadratic onset of ρ_{MB} without square-root behaviour on the evanescent axis, inside the contour. Within the Miller–Brown model the lateral wave could only be removed by switching roughness off at grazing angles

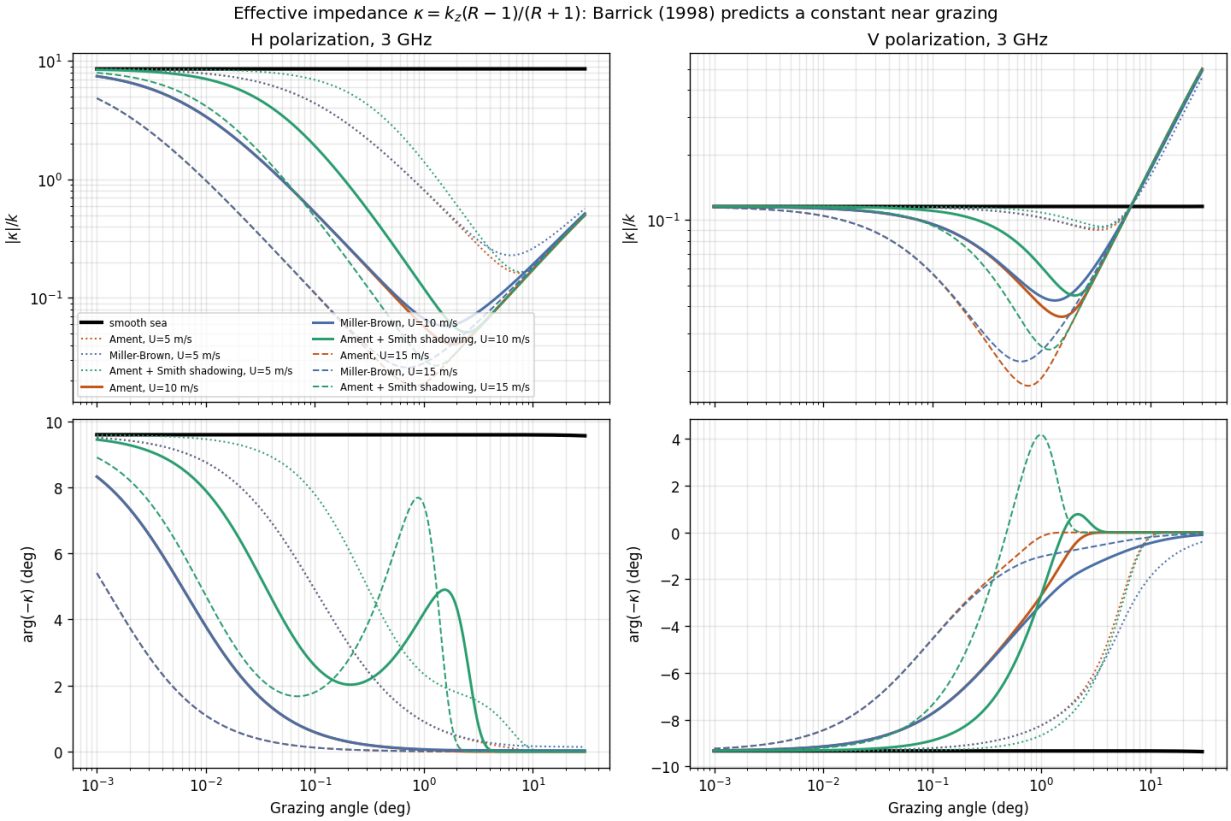


Figure 2: Effective impedance $\kappa = k_z(R-1)/(R+1)$ against the grazing angle, 3 GHz, $\varepsilon = 70$, 4 S/m, $\sigma = 0.0051U^2$; the Smith shadowing uses the Cox–Munk upwind slope. Top: $|\kappa|/k$; bottom: $\arg(-\kappa)$.

up to about 0.16° at X band, which are exactly the angles that matter for ducting. The beyond-horizon excess is therefore a consequence of extrapolating Miller–Brown to grazing angles of order 10^{-5} rad. The next section keeps the roughness at these angles but replaces the extrapolation by a constant impedance.

6 Constant grazing impedance

Blend. In terms of the admittance $Y = 1/\kappa = (R+1)/\big(k_z(R-1)\big)$, the roughness adds $Y_r = Y - Y_f$ to the smooth-sea admittance Y_f; near grazing Miller–Brown gives $Y_r \approx -\sigma^2 k_z$, the odd term of the previous section. We replace it by

$$Y_B(k_z) = Y_f(k_z) + Y_r\Big(\sqrt{k_z^2 + k_g^2}\Big), \qquad R_B = \frac{k_z Y_B + 1}{k_z Y_B - 1}, \tag{10}$$

where Y_r is evaluated with the exact Miller–Brown coefficient. Y_f and the argument of Y_r are even in k_z, so the impedance is even and there is no roughness-induced lateral wave. Below k_g the impedance is constant, $\kappa_g = 1/\big(Y_f(0) + Y_r(k_g)\big)$, as required by [8]; well above k_g the blend coincides with Miller–Brown. $R_B(0) = -1$, and $|R_B| \le 1$ on the real axis (checked numerically for 1–10 GHz and 5–15 m/s). The square root has branch points at $k_z = \pm i k_g$, so the blend is used with the zero-free fit only, which samples R on the real axis and on the sectors below it.

Blend angle. We place $k_g = 1/(2\sigma)$, where the Rayleigh roughness parameter $2k_z\sigma$ reaches one: Kirchhoff (Miller–Brown) above, effective impedance below. This is 0.89° at 3 GHz and 10 m/s and 0.40° at 15 m/s. For strong roughness $\kappa_g \approx -2/\sigma$, nearly real: $\kappa_g/k = -0.072$ at 3 GHz and 10 m/s, against $|\kappa|/k = 8.6$ for the smooth sea. The constant is a heuristic. Barrick's theory fixes

Table 2: Beyond-horizon decay (dB/km) in the standard atmosphere: PE slope at 10 m, 45–58 km, against the first creeping mode of the same boundary condition; last column: rough minus smooth path loss at 58 km. Antenna at 20 m, H-pol.

Band	Surface	PE	Mode 1	Excess loss (dB)
S	smooth	1.613	1.615	–
	Miller–Brown, 10 m/s	1.844	1.613	+2.7
	Miller–Brown, 15 m/s	0.048	1.605	−8.7
	blend, 10 m/s	1.595	1.596	−0.7
	blend, 15 m/s	1.566	1.573	−1.8
X	smooth	2.413	2.412	–
	Miller–Brown, 10 m/s	−0.020	2.398	−26.4
	Miller–Brown, 15 m/s	0.388	2.332	−39.7
	blend, 10 m/s	2.357	2.351	−2.4
	blend, 15 m/s	1.913	2.278	−11.3
	constant κ_g, 15 m/s	2.274	2.271	−5.1

that the impedance is constant, not its value. A self-consistent choice $k_g = |\kappa_g|$ (the constant region ending at the pseudo-Brewster angle) has no solution at H polarisation, where the whole angular range satisfies $\sin\psi < |\kappa|/k$, and at V polarisation returns the smooth-sea pseudo-Brewster angle. A calibrated value requires rigorous solutions over spectrum-generated sea surfaces, by the method of moments [7] or Monte Carlo PE [19].

Check against the modes. Beyond the horizon the field of a constant impedance is a sum of creeping modes with no cut contribution. With $u = \mathrm{Ai}\big(\omega(-z/\ell - t)\big)$, $\omega = e^{2\pi i/3}$, $\ell = (k^2 g)^{-1/3}$, the condition $u_z = i\kappa u$ gives $\mathrm{Ai}(s) + \omega\,\mathrm{Ai}'(s)/(i\kappa\ell) = 0$ with $s = -\omega t$, and the mode attenuation is $-\mathrm{Im}\, t/(2k\ell^2)$. For the blend the same equation is solved with $\kappa(k_z)$, $k_z = \sqrt{t}/\ell$. Table 2 compares the first-mode attenuation with the range slope of the PE path loss at 10 m over 45–58 km, less 0.084 dB/km of cylindrical spreading.

The PE follows the modes of the blend to 0.01 dB/km at S band and at X band, 10 m/s; the Miller–Brown runs do not, because the lateral wave dominates. The rough-sea field then stays within a few decibels of the smooth one: the small, nearly real κ_g makes the surface less Dirichlet-like than the smooth sea and lowers the creeping-mode attenuation by 0.02–0.13 dB/km. At X band and 15 m/s the blend angle lies inside the creeping-mode range, $k_g = 0.44 < 3(k^2 g)^{1/3} = 0.72\,\mathrm{m}^{-1}$. The modes then sample the transition of the blend and its branch point, and the far field depends on the blend shape: the slope is 1.91 dB/km with the Rayleigh k_g, 2.18 dB/km with $k_g = 1/\sigma$, and 2.27 dB/km for a constant κ_g at all angles, which again matches its mode. The implementation warns in this case. In the 15 m evaporation duct at S band both models change the loss at 10 m by less than 1.5 dB over 60 km (Fig. 3).

7 Results

7.1 Validation against an exact solution

For a Gaussian beam over a flat surface in a homogeneous medium the one-way field is the plane-wave spectrum of the source, each down-going component reflected with $R(|k_z|)$. Table 3 lists the relative error of $|u|$ beyond the specular point. Short-range cases use 3 GHz, a 30 m antenna and 5° elevation; X band cases use 5 km, a 20 m antenna and 0.5° elevation. The smooth-sea error is 0.6 %.

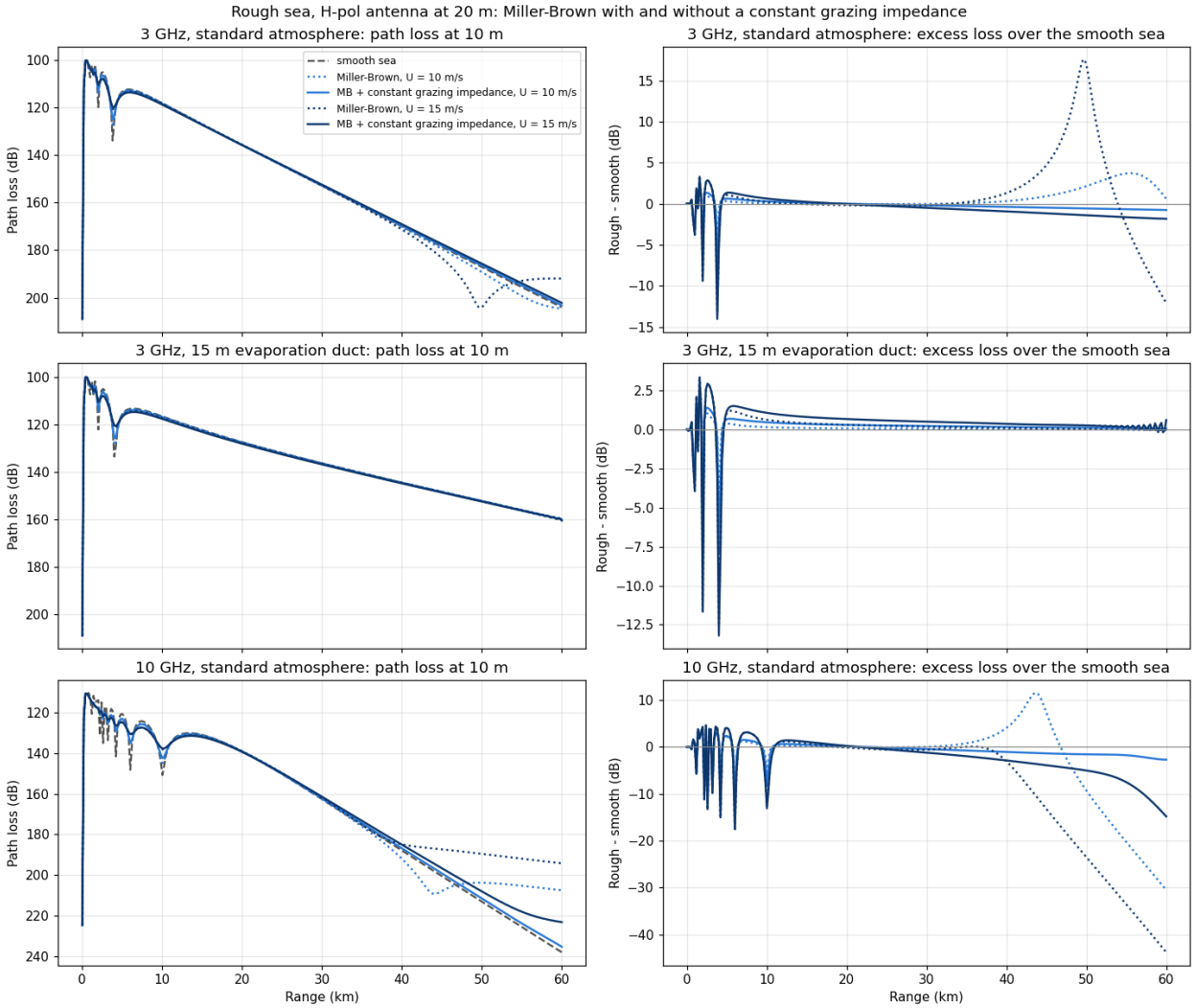


Figure 3: Path loss at 10 m (left) and excess over the smooth sea (right): Miller–Brown (dotted) and Miller–Brown with the constant grazing impedance (solid), 10 and 15 m/s. Rows: S band in a standard atmosphere and in a 15 m evaporation duct, X band in a standard atmosphere.

7.2 Long-range demonstration

A 20 m H-pol antenna with a 2° beam radiates over 60 km of sea ($\varepsilon = 70$, 4 S/m) at 1, 3 and 10 GHz, for wind speeds of 0, 5, 10 and 15 m/s and three refractivity profiles: standard atmosphere, a 15 m evaporation duct, and a surface-based duct with a 30 m trapping layer at 100 m. All 36 cases are stable. Figures 5–6 show the results, and Table 4 gives the roughness-induced loss.

Discussion. In the evaporation duct at X band the roughness-induced loss grows steadily with range, by about 0.1 dB/km at 15 m/s, reaching 6.4 dB at 58 km. The ducted field is carried by the lowest trapped modes, which meet the surface at about 0.1°. There the Miller–Brown factor is 0.73 at 15 m/s, i.e. −2.8 dB per bounce with a bounce every 17 km or so: about 0.16 dB/km, the same order as the computed rate. At 60 km the effect is moderate; if the rate persists, it would reach 10–15 dB over 100–150 km. At S and L band roughness is negligible in the ducts, since $k\sigma \sin\psi$ stays small at the grazing angles of the trapped energy. In the surface-based duct, the energy trapped below the layer at 100 m interacts with the surface less often; roughness smooths the interference pattern below about 60 m at X band (Fig. 5) and adds up to 5 dB of loss. Within the horizon in the standard atmosphere roughness changes the loss by less than 1 dB. Beyond the horizon the Miller–Brown lateral wave dominates. The fitted coefficient needed lifts of 0–0.86 m in

Table 3: Relative error of $|u|$ against the exact solution, zero-free fit.

Frequency	σ	Lift z_0	H-pol	V-pol
3 GHz	0.1 m	0–0.01 m	0.8 %	1.2 %
3 GHz	0.3 m	0–0.03 m	1.1 %	1.1 %
10 GHz	0.51 m	0.10 m	0.08 %	0.15 %
10 GHz	1.15 m	0.35–0.58 m	0.16 %	0.09 %
constant grazing impedance (Section 6), exact R_B in the reference				
3 GHz	0.3 m	0	1.1 %	1.1 %
10 GHz	0.51 m	0	0.13 %	0.15 %

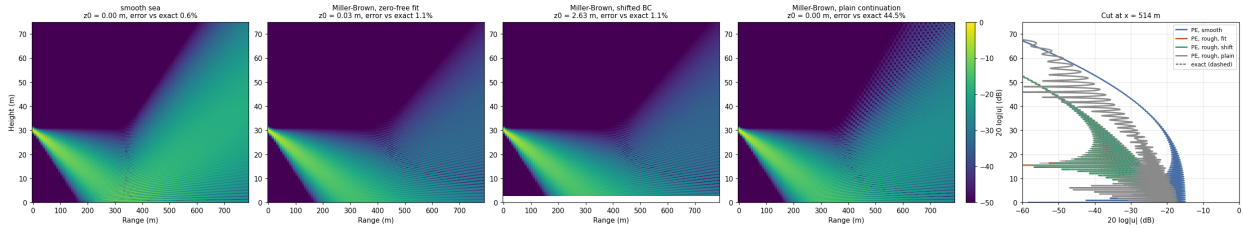


Figure 4: 3 GHz H-pol beam at 5° grazing over a sea with $\sigma = 0.3$ m: smooth sea, zero-free fit, lifted reference plane, plain continuation, and a vertical cut through the reflected beam against the exact solution (dashed).

all cases except X band at 15 m/s in the ducts (1.72 m), against 15–78 m for the pure lift. Section 6 revisits the beyond-horizon cases with the constant grazing impedance.

Computational cost. A rough-sea model requires a probe of the grid (no lower NLBC), the contour, a few least-squares fits, and the lower NLBC. Builds take 10–65 s at L and S band and about 3 min at X band over 60 km (five models in parallel on 20 cores). The constant grazing impedance builds about twice as fast at S band and needs smaller lifts (none in the standard atmosphere at S band). When several models are built in parallel processes, the BLAS threads of each process must be limited. With four processes on 20 cores and default thread pools, OpenBLAS busy-waiting made the fits about 20 times slower (1430 s against 67 s per model).

8 Conclusions

The discrete NLBC handles a rough sea once the reflection coefficient is given a suitable continuation. The plain continuation of the Miller–Brown factor is unstable, and no bounded continuation can be exact near grazing. A zero-free fitted coefficient $1+\tilde{R} = (1+R_f)e^K$, with at most a small lift of the reference plane, reproduces Miller–Brown to about 1 %. Its stability on long-range problems requires, beyond zero-freeness, passivity for propagating waves and in the trapping range of the atmosphere, and accuracy on the wavenumbers of range-decaying modes. In ducts the expected roughness-induced attenuation that grows with range is obtained at X band, and roughness is negligible at L band. Beyond the horizon Miller–Brown raises the field substantially; the comparison with an independent modal and branch-cut solution identifies this as the lateral wave of the model at grazing incidence, where its effective impedance behaves like $1/k_z$. This contradicts the constant grazing impedance of rough surfaces [8]. A blend that freezes the Miller–Brown roughness admittance below the Rayleigh angle removes the lateral wave, and with the fit matched to first order at grazing the PE reproduces the creeping modes of the resulting impedance to 0.01 dB/km. The value of the grazing impedance remains to be calibrated against rigorous solutions over sea surfaces generated from wave spectra; the boundary condition accepts any such calibrated coefficient.

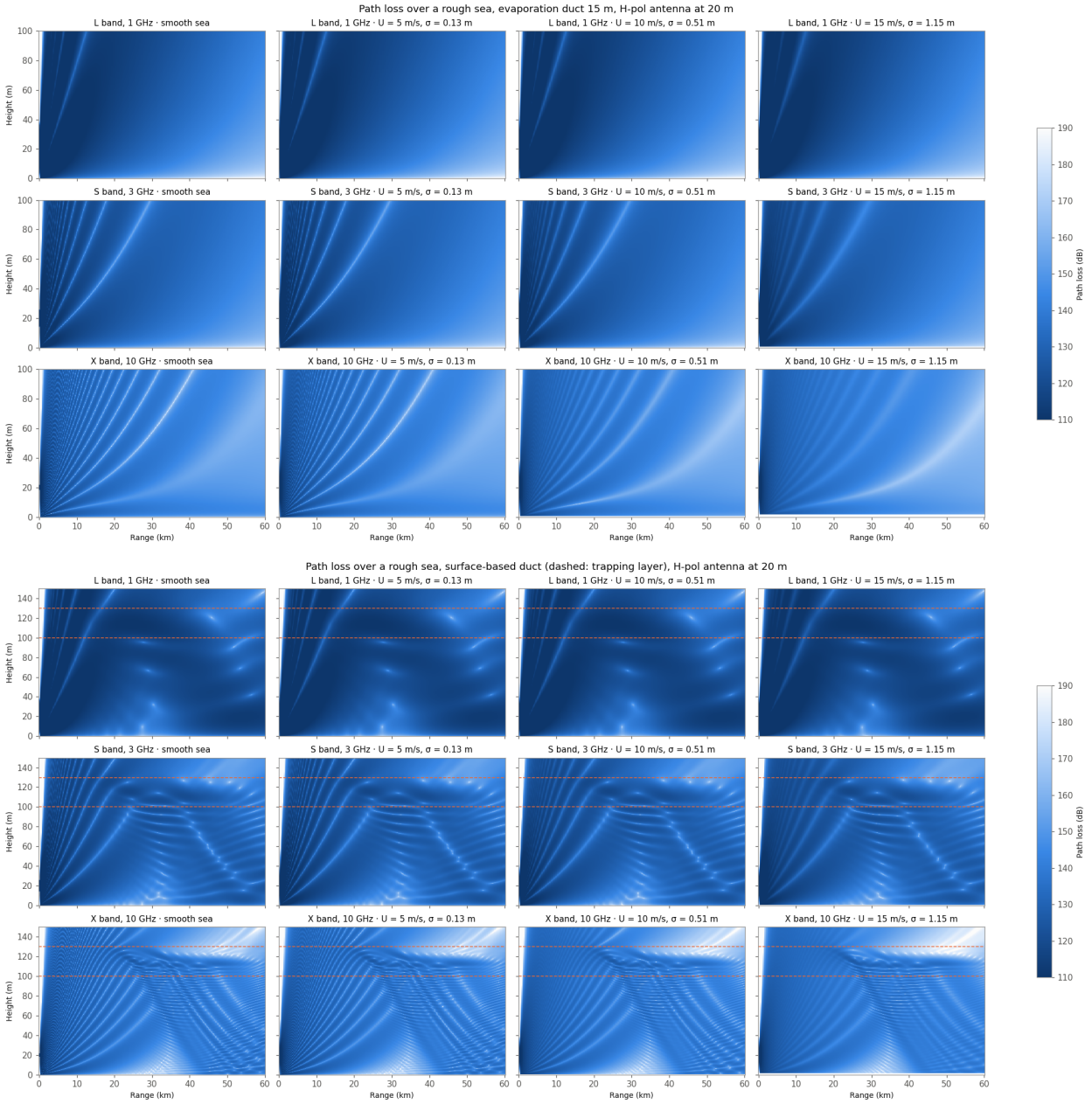


Figure 5: Path loss over a rough sea: 15 m evaporation duct (top) and surface-based duct (bottom, dashed: trapping layer). Rows: L, S, X band; columns: smooth sea and 5, 10, 15 m/s.

Table 4: Roughness-induced excess path loss (dB) at 10 m height: mean over 20–60 km / value at 58 km, for wind speeds 5, 10 and 15 m/s ($\sigma = 0.13$, 0.51, 1.15 m). Negative values beyond the horizon in the standard atmosphere are the Miller–Brown lateral wave (Section 5).

Profile	Band	5 m/s	10 m/s	15 m/s
Standard	L	$+0.0/+0.0$	$+0.2/+0.7$	$+0.7/+3.7$
	S	$+0.1/+0.2$	$+1.0/+2.8$	$+1.3/-8.7$
	X	$-0.9/-7.0$	$-4.0/-26.4$	$-12.0/-39.7$
Evaporation duct 15 m	L	0.0/0.0	0.0/0.0	$0.0/-0.1$
	S	0.0/0.0	0.0/0.0	+0.1/0.0
	X	$+0.2/+0.3$	$+2.1/+2.6$	$+4.5/+6.4$
Surface-based duct	L	0.0/0.0	$+0.1/+0.1$	$+0.1/+0.2$
	S	0.0/0.0	$0.0/+0.3$	$+0.1/+1.1$
	X	$0.0/+0.1$	$+0.5/+1.8$	$+0.8/+4.9$

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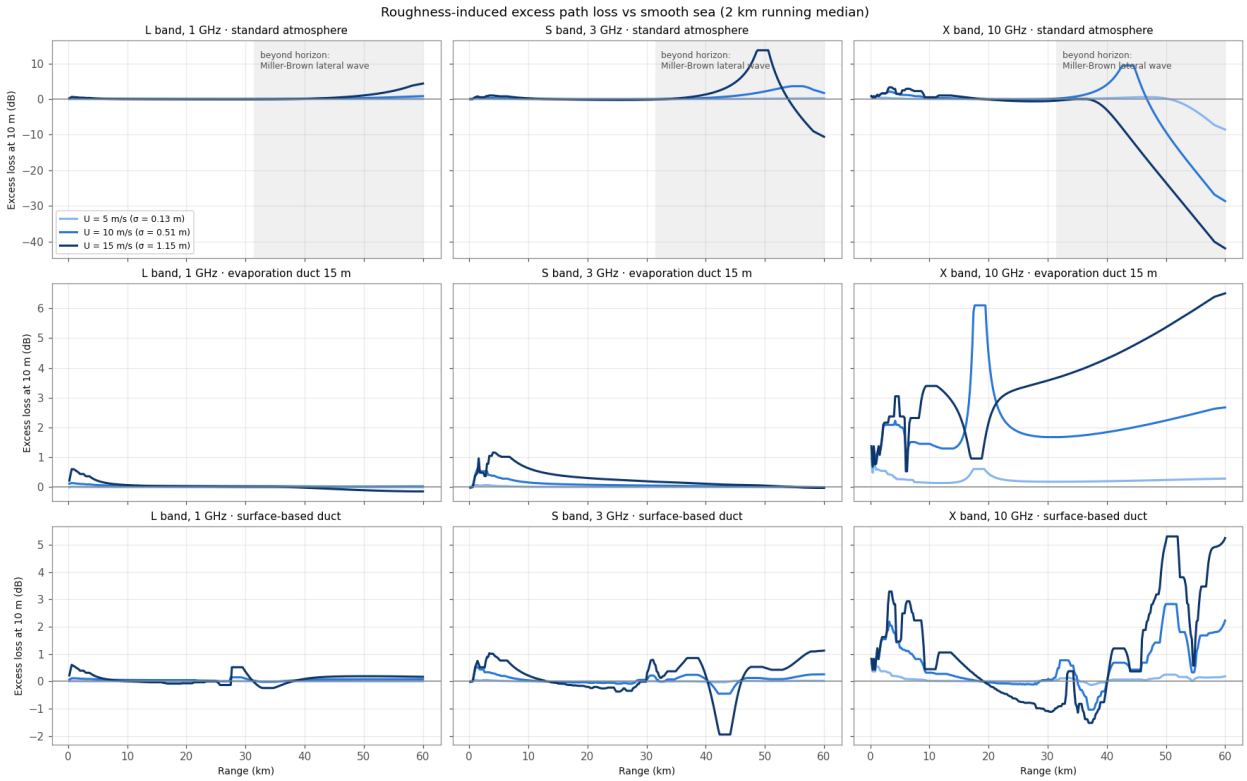


Figure 6: Roughness-induced excess path loss at 10 m height vs range (2 km running median). Shaded: beyond the radio horizon in the standard atmosphere, where the Miller–Brown lateral wave dominates (Section 5).

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